Reactor core vessel Specification

Document number	SHV-HR-SPEC-0002
Revision	D
Status	APPROVED
Classification	CONFIDENTIAL
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Approved by	Aldric Voss
Date	2025-01-19

1. Purpose

This specification defines the performance, interface and verification requirements for the reactor core vessel (HR-RCV) of the Hypermatter Reactor on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-HR-02 (WBS 1.1.2), owned by Jessa Pell.

The prime supplier for flight hardware is Mon Gazza Transport. Responsible engineer: Elara Drummond.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-HR-DWG-0201	Reactor core vessel general arrangement drawing
SHV-HR-BOM-0301	Reactor core vessel bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-HR-TX	Interface Control Document: Hypermatter Reactor / Thermal Control & Heat Rejection
SHV-ICD-HR-PD	Interface Control Document: Hypermatter Reactor / Power Distribution
SHV-ICD-HR-CN	Interface Control Document: Hypermatter Reactor / Computer Core & Data Network

4. Description

The hypermatter reactor is the station's main power plant. It supplies every zone substation through the power distribution system and is backed by the solar collector array.

The reactor core vessel is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
HR-RCV-R01	The reactor core vessel shall provide a core diameter of 1.85 km.	1.85 km	±0.5%	Inspection
HR-RCV-R02	The reactor core vessel shall provide a wall thickness of 18.7 m.	18.7 m	±0.5%	Demonstration

ID	Requirement	Value	Tolerance	Verification
HR-RCV-R03	The reactor core vessel shall provide a design core temperature of 4,400 K.	4,400 K	±0.5%	Test

5.1 General requirements

ID	Requirement	Verification
HR-RCV-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
HR-RCV-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
HR-RCV-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
HR-RCV-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
HR-RCV-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the HR test-report series.

8. Revision history

Rev	Date	Description	Author
A	2023-08-16	Initial release.	Elara Drummond
B	2023-12-26	Clarified verification methods.	Elara Drummond
C	2024-06-27	Incorporated interface changes.	Elara Drummond
D	2025-01-19	Design core temperature set to 4,400 K (approved).	Elara Drummond